Econometrics I

Lecture 8a: Panel Data and Fixed Effects

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Where we are

- Weeks 5–7: identification by design — randomize, or find an instrument.
- Today: identification by repeated observation. We see the same firm, store, customer, or state many times. What does that buy us?
  - Part a (this deck): fixed effects, random effects, and the choice between them.
  - Part b: event studies — the panel version, and the earnings-announcement version.
- The one idea: things about unit i that *never change* can be removed without ever measuring them.
- Readings: Hansen Ch. 17; Mixtape Ch. 8.

What a panel is

- Units $i = 1, \dots, N$ observed at periods $t = 1, \dots, T$ (or T_i if unbalanced).
- The panels you will meet are wide and short: N large, T small.
  - firms $\times$ fiscal years (Compustat: $N \approx 5{,}000$, $T \approx 10$–40)
  - stores $\times$ weeks (scanner data)
  - customers $\times$ months (subscription, churn)
  - hospitals or physicians $\times$ quarters
- Everything today is "$N \to \infty$, T fixed." Large-T panels (countries, 50 years) are a different world: unit roots, cointegration, Nickell bias that goes away.
- Notation: y_{it} outcome, x_{it} a k-vector of regressors. Bars are unit means: $\bar{y}_i = T^{-1} \sum_t y_{it}$.

The model

$$y_{it} = x_{it}'\beta + \alpha_i + \varepsilon_{it}$$

- α_i: everything about unit i that does not change over t. Management quality, brand, location, a CEO's risk appetite, a customer's taste.
- ε_{it}: the idiosyncratic shock. Different every period.
- The conditional mean we want is $\mathbb{E}[y_{it} \mid x_{it}, \alpha_i] = x_{it}'\beta + \alpha_i$. The whole subject: what happens when you *cannot* condition on α_i?
- Two names for α_i, and they are not synonyms:
  - a fixed effect: treat α_i as a parameter (or a nuisance) — no assumption about how it relates to x_{it}.
  - a random effect: treat α_i as a draw, *uncorrelated* with x_{it}.

Running example: management practices and productivity

- Bloom and Van Reenen (2007) score firms on management practices (monitoring, targets, incentives) and ask whether better practices raise productivity.
- The obvious worry: well-run firms adopt good practices *and* are productive for other reasons. Firm quality α_i drives both.
- Our simulated version, with ground truth:

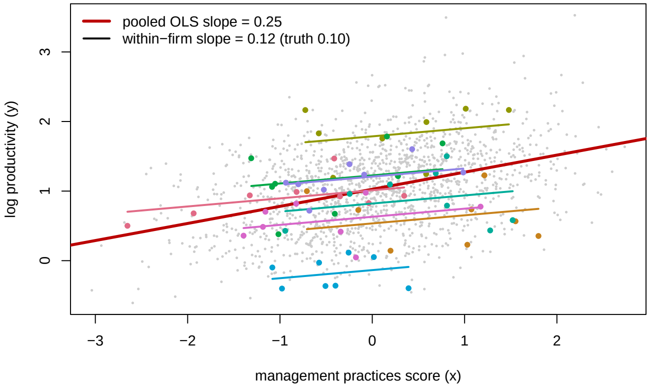
$$y_{it} = 1 + 0.10\, x_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \qquad \alpha_i \sim N(0, 0.5^2),\ \varepsilon_{it} \sim N(0, 0.3^2)$$

$$x_{it} = 0.7\, \alpha_i + \text{(persistent firm component)} + \text{(year-to-year noise)}$$

$N = 200$ firms, $T = 8$ years. y is log productivity, x the practices score.

- We know $\beta = 0.10$. Every estimator gets judged against that number.

The picture



Grey: all firm-years. Colored: eight firms, each with its own within-firm fit. The pooled line runs through the *between-firm* cloud; the within-firm lines have the true slope.

Pooled OLS: when is it fine?

Ignore the panel structure and regress y_{it} on x_{it} (and a constant).

$$\hat{\beta}_{\text{pooled}} \xrightarrow{p} \beta + \frac{\mathbb{C}(x_{it}, \alpha_i)}{\mathbb{V}(x_{it})}$$

- Same omitted-variable formula as Week 2. The omitted variable is α_i.
- Consistent iff $\mathbb{C}(x_{it}, \alpha_i) = 0$ — the "random effects" assumption.
- In the example, $\mathbb{C}(x, \alpha) > 0$: pooled OLS is biased *up*. Good practices look more effective than they are because good firms have them.
- Even when unbiased, pooled OLS standard errors are wrong: α_i sits in the error of every period for firm i, so errors are correlated within firm. Cluster by firm (Week 3).

Three ways to get rid of α_i

1. Dummies (LSDV). Put in N firm indicators and run OLS. Fine for $N = 200$; awkward for $N = 5$ million customers.

2. Within transformation. Subtract the firm mean from every variable:

$$(y_{it} - \bar{y}_i) = (x_{it} - \bar{x}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

   $\alpha_i - \bar{\alpha}_i = 0$. It is gone. OLS on the demeaned data is the fixed-effects (FE) estimator.

3. First differences. $\Delta y_{it} = \Delta x_{it}'\beta + \Delta\varepsilon_{it}$. Also kills α_i.

- (1) and (2) give *identical* $\widehat{\beta}$ (FWL: demeaning is projecting off the dummies). (3) equals them when $T = 2$; differs otherwise.
- All three use only within-firm variation. That is the price and the point.

The within transformation as a projection

- Stack firm i's T observations. Let ι be a T-vector of ones and

$$M = I_T - \tfrac{1}{T}\iota\iota' \qquad \text{so that} \qquad My_i = y_i - \bar{y}_i\iota.$$

  M is symmetric and idempotent: it is the residual-maker from regressing on a constant.

- The FE estimator is OLS after applying M firm by firm:

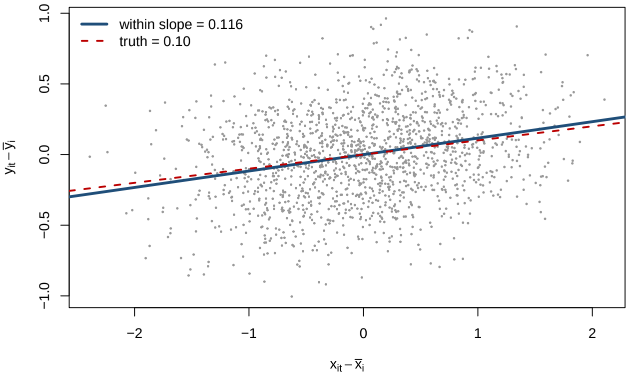
$$\hat{\beta}_{FE} = \Big(\sum_i x_i' M x_i\Big)^{-1} \sum_i x_i' M y_i = \Big(\sum_i \sum_t \tilde{x}_{it}\tilde{x}_{it}'\Big)^{-1} \sum_i \sum_t \tilde{x}_{it}\tilde{y}_{it}$$

  with $\tilde{x}_{it} = x_{it} - \bar{x}_i$.

- This is Frisch–Waugh–Lovell from Week 2 with N dummies as the “other” regressors. Software (`fixest`, `pyfixest`) does the demeaning without ever forming the dummies.

- Degrees of freedom: you spent N of them. $\hat{\sigma}^2_\varepsilon$ divides by $NT - N - k$.

The within picture



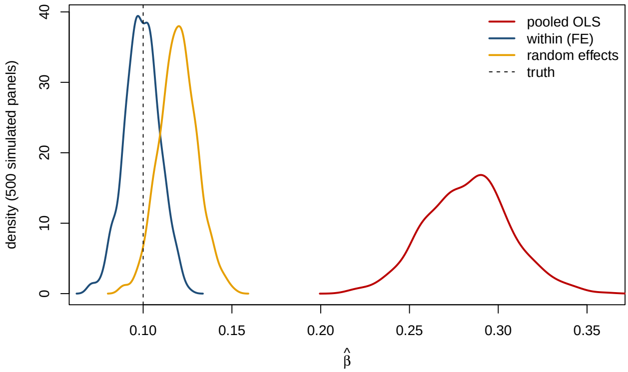
After demeaning, every firm's cloud is centered at the origin. The between-firm variation that carried the bias is gone; what is left is the true slope, plus noise.

Estimates on the simulated panel

	Pooled OLS	First diff.	Within (FE)	Random eff.	Mundlak
$\hat{\beta}$ (truth $= 0.10$)	0.246	0.105	0.116	0.127	0.116
SE (cluster firm)	(0.023)	(0.013)	(0.011)	(0.011)	(0.011)
Coef. on $\bar{x}_i$					0.376 (0.054)
Firm FE	no	(differenced)	yes	quasi ($\hat{\theta} = 0.72$)	no
$N \times T$	1600	1400	1600	1600	1600

- Pooled OLS is off by a factor of two and a half. FE and first differences are within sampling error of 0.10 (one draw; the next slide shows the distribution).
- Random effects and Mundlak are coming. Note the Mundlak coefficient on $\bar{x}_i$: it is the part of the pooled bias that FE removed, estimated directly.

Sampling distributions (500 simulated panels)



The pooled estimator is precisely wrong. FE is centered on the truth. RE is in between: it is a weighted average of within and between information, and the between part is contaminated.

Strict exogeneity: the assumption you actually need

For FE to be unbiased we need

$$\mathbb{E}[\varepsilon_{it} \mid x_{i1}, \dots, x_{iT}, \alpha_i] = 0 \qquad \text{for every } t.$$

- Stronger than $\mathbb{E}[\varepsilon_{it} \mid x_{it}] = 0$: today's shock must be unrelated to *past and future* x. Why? Demeaning mixes every period's x into every period's regressor.
- What it rules out: feedback. A firm that has a bad year ($\varepsilon_{it} < 0$) and responds by adopting new practices ($x_{i,t+1} \uparrow$) violates it. A store that sees demand jump and raises prices next week violates it.
- What it allows: α_i correlated with x in any way at all. That is the whole appeal.
- First differences need only $\mathbb{E}[\Delta\varepsilon_{it} \mid \Delta x_{it}] = 0$, which is weaker in one direction and stronger in another; the two estimators differing a lot is itself a diagnostic.

What FE cannot do

- Time-invariant regressors drop out. Industry, founding year, headquarters state: all are absorbed by α_i. FE cannot tell you their effect; it *controls* for them.
- Identification comes from movers only. If practices never change within a firm, $\tilde{x}_{it} = 0$ and the firm contributes nothing. Ask: who changes x, and why?
- Measurement error bites harder. Demeaning removes the persistent (signal) part of x and keeps the noisy part. Attenuation grows:

SD of measurement error in x	0	0.25	0.5	0.75	1.0
Pooled OLS ($\rho = 0$, so unbiased without error)	0.100	0.091	0.080	0.066	0.048
Within (FE)	0.100	0.092	0.071	0.054	0.038

(Here $\rho = 0$, so pooled OLS is unbiased without noise; the point is the comparison.)

- Precision. You are using less variation. Standard errors go up.

Two-way fixed effects

$$y_{it} = x_{it}'\beta + \alpha_i + \gamma_t + \varepsilon_{it}$$

- γ_t: shocks common to everyone in period t. The business cycle, a platform-wide price change, seasonality, a pandemic.
- Estimation: demean by unit *and* by period (iterate until both means are zero, which is what `fixest` does). Equivalent to unit dummies plus period dummies.
- Interpretation of β: the effect of a change in x_{it} *relative to firm i's usual level and relative to what happened to everyone else in year t.*
- The workhorse of Weeks 8–9: almost every difference-in-differences you will read is a two-way FE regression.

Random effects: the model as variance components

Suppose α_i is a draw, $\alpha_i \sim (0, \sigma_\alpha^2)$, independent of x and ε. Write the composite error $v_{it} = \alpha_i + \varepsilon_{it}$. Then, for firm i:

$$\mathbb{V}(v_{it}) = \sigma_\alpha^2 + \sigma_\varepsilon^2, \qquad \mathbb{C}(v_{it}, v_{is}) = \sigma_\alpha^2 \;\; (t \neq s), \qquad \text{Corr}(v_{it}, v_{is}) = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_\varepsilon^2} \equiv \rho.$$

- Every pair of periods within a firm is equicorrelated with correlation ρ. Stacked, $\Omega_i = \mathbb{V}(v_i) = \sigma_\varepsilon^2 I_T + \sigma_\alpha^2 \iota\iota'$.
- Pooled OLS is unbiased under this model but inefficient: it ignores that we have "N clusters of T correlated observations," not NT independent ones.
- The efficient estimator is GLS with this Ω. That is what random effects means.

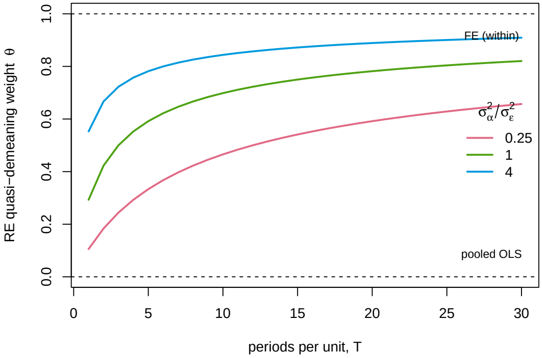
Random effects as quasi-demeaning

GLS with $\Omega_i = \sigma_\varepsilon^2 I_T + \sigma_\alpha^2 \iota\iota'$ turns out to be OLS on *partially* demeaned data:

$$(y_{it} - \theta\bar{y}_i) = (x_{it} - \theta\bar{x}_i)'\beta + (v_{it} - \theta\bar{v}_i), \qquad \theta = 1 - \sqrt{\frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + T\sigma_\alpha^2}}.$$

- $\theta = 0$: pooled OLS. $\theta = 1$: fixed effects. RE sits in between, and the position is chosen by the data: how much of the variance is α, and how many periods you have.
- Read the formula: as $T \to \infty$, $\theta \to 1$ and RE *becomes* FE. With many periods per unit, the unit mean is estimated so well that GLS just removes it.
- As $\sigma_\alpha^2 \to 0$, $\theta \to 0$: no firm heterogeneity, nothing to remove.
- Sketch of why: $\Omega_i^{-1/2}$ for an equicorrelated matrix is $\sigma_\varepsilon^{-1}\left[I_T - \theta\,\frac{1}{T}\iota\iota'\right]$, and applying it is exactly the quasi-demeaning above.

How much demeaning? θ as a function of T



For a typical firm panel ($\sigma^2_\alpha \geq \sigma^2_\varepsilon, T \approx 10$), θ is already above 0.8. RE and FE will be close in *value* whenever they are both consistent; the question is only whether RE is consistent.

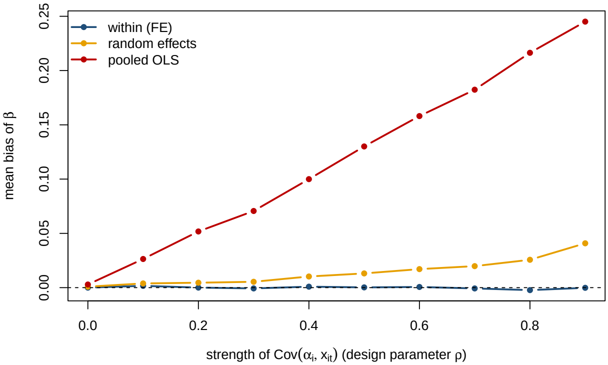
Feasible GLS: estimating the variance components

θ depends on σ^2_ε and σ^2_α, which we estimate first (Swamy–Arora):

1. Run the within regression. Its residual variance estimates σ^2_ε (the α_i are gone, so only ε is left): $\hat{\sigma}^2_\varepsilon = \frac{\sum_{i,t} \tilde{e}^2_{it}}{NT-N-k}$.
2. Run the between regression: $\bar{y}_i$ on $\bar{x}_i$, one observation per firm. Its residual variance estimates $\sigma^2_\alpha + \sigma^2_\varepsilon/T$. Subtract: $\hat{\sigma}^2_\alpha = \hat{\sigma}^2_{\text{between}} - \hat{\sigma}^2_\varepsilon/T$ (truncate at zero).
3. Form $\hat{\theta}$, quasi-demean, run OLS.

- The within estimator uses only within variation; the between estimator uses only between variation. RE is the precision-weighted combination of the two.
- That sentence is also the diagnosis: if the between variation is contaminated by $\mathbb{C}(\bar{x}_i, \alpha_i) \neq 0$, RE inherits a share $(1-\theta)$-ish of the pooled bias.
- The workbook implements this in a dozen lines of R; `plm` and `lme4` do it for you.

RE bias as the correlation between α_i and x_{it} grows



FE does not care. RE is biased in proportion to the correlation, less than pooled OLS because θ is large here ($T = 8$), but biased. Nothing about “random” protects you from this.

Fixed vs. random: what the words actually mean

- Both treat α_i as unobserved and ε_{it} as random. The difference is *one assumption*:

$$\text{RE: } \mathbb{E}[\alpha_i \mid x_{i1}, \dots, x_{iT}] = 0 \qquad\qquad \text{FE: no restriction on } \mathbb{E}[\alpha_i \mid x_{i1}, \dots, x_{iT}]$$

- Under the RE assumption both are consistent and RE is more efficient; under its failure only FE is consistent. Bias beats variance: economists default to FE. (Hansen: the distinction is about what you assume, not whether α_i is "really" random.)
- Why anyone runs RE anyway:
  - you need the coefficient on a time-invariant regressor (industry, gender, region);
  - you believe the assumption because x was randomized across units;
  - you care about α_i itself and want to shrink noisy unit means (teacher effects, hospital quality, seller ratings) — "empirical Bayes" is RE by another name;
  - your field calls it a hierarchical or mixed model and everyone uses it.

The Hausman test

Under the RE assumption, $\widehat{\beta}_{FE}$ and $\widehat{\beta}_{RE}$ estimate the same thing; if it fails, they diverge. So test whether they differ:

$$H = (\widehat{\beta}_{FE} - \widehat{\beta}_{RE})' \left[\widehat{V}_{FE} - \widehat{V}_{RE}\right]^{-1} (\widehat{\beta}_{FE} - \widehat{\beta}_{RE}) \xrightarrow{d} \chi^2_k \quad \text{under } H_0.$$

- The variance of the difference collapses to $V_{FE} - V_{RE}$ because RE is efficient under H_0 (the efficient estimator is uncorrelated with the difference). Neat, but fragile:
  - it requires the *classical* RE variance — no clustering, no heteroskedasticity;
  - $\widehat{V}_{FE} - \widehat{V}_{RE}$ need not be positive definite in finite samples;
  - only regressors that vary within unit can be compared.
- Rejecting says "RE is inconsistent, use FE." Not rejecting says "you could not tell": with few periods RE and FE are noisy and the test has little power.
- There is a better way to run the same test.

Mundlak: put $\bar{x}_i$ in the regression

Model the correlation instead of assuming it away. Let

$$\alpha_i = \bar{x}_i'\gamma + u_i, \qquad \mathbb{E}[u_i \mid x_{i1}, \dots, x_{iT}] = 0.$$

Substitute:

$$y_{it} = x_{it}'\beta + \bar{x}_i'\gamma + u_i + \varepsilon_{it}.$$

- Now u_i satisfies the RE assumption by construction. Run RE (or even pooled OLS) on $(x_{it}, \bar{x}_i)$.
- Two facts (Mundlak 1978; the correlated random effects model): the coefficient on x_{it} is *exactly* $\hat{\beta}_{FE}$, and $\hat{\gamma}$ is exactly the between-minus-within difference. The Hausman test is $H_0 : \gamma = 0$.
- Advantages over the classical Hausman statistic: cluster-robust standard errors are fine; nothing needs to be positive definite; you can keep time-invariant regressors in the model.

Test	Statistic	p-value	Verdict
Hausman $(\widehat{\beta}_{FE} - \widehat{\beta}_{RE})^2/(\widehat{V}_{FE} - \widehat{V}_{RE})$	-4.7	—	undefined: $\widehat{V}_{FE} < \widehat{V}_{RE}$
Mundlak t-test on $\bar{x}_i$ (cluster-robust)	6.9	< 0.001	reject RE

- The classical statistic is *negative*: $\widehat{V}_{FE} - \widehat{V}_{RE}$ is not positive definite in this sample, exactly the fragility from the previous slide. (RE is inconsistent here, so its residual variance is inflated by the bias, and its "efficient" standard error is not smaller than FE's.) Software will either report nonsense or refuse.
- The Mundlak version has no such problem, rejects decisively, and came with clustered standard errors for free.
- Look back at the estimates table: the Mundlak coefficient on x_{it} matched FE to three decimals.

Inference: cluster by the unit

- After demeaning, the errors $\varepsilon_{it} - \bar{\varepsilon}_i$ are correlated within firm mechanically, and ε_{it} itself is usually serially correlated too.
- Week 3's answer applies unchanged: cluster at the firm. Asymptotics are in N, the number of clusters, which is why "N large, T small" is the comfortable case.
- Bertrand, Duflo and Mullainathan (2004): in state-year panels with serially correlated outcomes, unclustered two-way FE regressions reject a true null of *no effect* 45% of the time at the 5% level. Cluster.
- Cluster at the level of the fixed effect *or* the level at which x is assigned, whichever is coarser. Firm FE with a state-level policy: cluster by state.
- Few clusters (under ≈ 40) is a separate problem; wild cluster bootstrap (Week 3).

$$y_{it} = \rho\, y_{i,t-1} + x_{it}'\beta + \alpha_i + \varepsilon_{it}$$

- Tempting: "control for last year's productivity." Demean it:

$$(y_{it} - \bar{y}_i) = \rho\,(y_{i,t-1} - \bar{y}_i) + \cdots + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

- $\bar{y}_i$ contains y_{it}, which contains ε_{it}; and $\bar{\varepsilon}_i$ contains $\varepsilon_{i,t-1}$, which is in $y_{i,t-1}$. Regressor and error are correlated by construction.
- The bias is $O(1/T)$: it does not vanish as $N \to \infty$. With $T = 8$ and $\rho = 0.5$, $\hat{\rho}$ is biased down by roughly 0.15, and $\hat{\beta}$ picks up some of the damage.
- Fixes (Anderson–Hsiao, Arellano–Bond) instrument $\Delta y_{i,t-1}$ with deeper lags. Rule for this course: no lagged dependent variable in an FE regression unless you know why.

Pitfall: when α_i is the object of interest

- Sometimes the fixed effect *is* the estimand: which physicians have better outcomes, which managers add value, which sellers deliver on time.
- $\hat{\alpha}_i = \bar{y}_i - \bar{x}_i'\hat{\beta}$ is unbiased but noisy with variance $\approx \sigma_\varepsilon^2/T_i$. The units at the top of the ranking are disproportionately the ones with small T_i and lucky draws.
- Shrink toward the mean in proportion to noise (empirical Bayes; the RE model again):

$$\hat{\alpha}_i^{EB} = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_\varepsilon^2/T_i}\,\hat{\alpha}_i.$$

- Believing $\hat{\alpha}_i$ causally also needs strict exogeneity of the *assignment* of units to i: patients are not randomly assigned to physicians (Chetty–Hendren's movers design is the fix).
- Nonlinear models (logit with firm FE) add the incidental parameters problem: $\hat{\beta}$ is inconsistent for fixed T. Week 12.

In practice

R (`fixest`):

```
feols(y ~ x | firm + year, data = d, cluster = ~firm)          # two-way FE
d[, xbar := mean(x), by = firm]
feols(y ~ x + xbar | year, data = d, cluster = ~firm)           # Mundlak / CRE
```

Python (`pyfixest`):

```
pf.feols("y ~ x | firm + year", data=d, vcov={"CRV1": "firm"})
d["xbar"] = d.groupby("firm")["x"].transform("mean")
pf.feols("y ~ x + xbar | year", data=d, vcov={"CRV1": "firm"})
```

- Random effects proper: `plm(..., model = "random")` or `lme4::lmer(y ~ x + (1|firm))`; `statsmodels.MixedLM` in Python. The workbook writes FGLS by hand so you see $\widehat{\theta}$.

- α_i is everything time-invariant about unit i. Pooled OLS is biased by $\mathbb{C}(x, \alpha)/\mathbb{V}(x)$.
- FE removes α_i by demeaning (or dummies, or differencing) and needs strict exogeneity: no feedback from shocks to future x. It uses within variation only.
- RE is GLS with equicorrelated errors, i.e. quasi-demeaning with weight θ. It needs $\alpha_i \perp x$, and is biased otherwise.
- Mundlak: add $\bar{x}_i$. You get FE's slope, a robust Hausman test, and time-invariant regressors, all at once.
- Cluster by unit. Do not lag the outcome. Two-way FE is the template for next hour and next week.

`inclass_session8.Rmd` (and the `pyfixest` twin):

1. Reproduce the estimates table: pooled, within, FD, hand-rolled RE, Mundlak.
2. Turn ρ down to zero and up to 0.9. Watch RE move and FE stay put.
3. Add measurement error to x. Which estimator suffers more, and why?